

IN4MATX 232: Research in HCI

Class 1:
History & Course Overview

Daniel Epstein

Today's class

- Defining Human-Computer Interaction (HCI)
- History of HCI
- Course overview

Discussion time! Split into groups of 2-3.

What is Human-Computer Interaction?

Creating tomorrow together?



<https://chi2026.acm.org/>

Design and use of computer technology?

Human–computer interaction

From Wikipedia, the free encyclopedia

(Redirected from [Human-computer interaction](#))



This article **may contain an excessive amount of intricate detail that may interest only a particular audience**. Please help by [spinning off](#) or [relocating](#) any relevant information, and removing excessive detail that may be against [Wikipedia's inclusion policy](#). (August 2021) ([Learn how and when to remove this template message](#))

Human–computer interaction (HCI) is research in the design and the use of [computer technology](#), which focuses on the [interfaces](#) between people ([users](#)) and [computers](#). HCI researchers observe the ways humans interact with computers and design technologies that allow humans to interact with computers in novel ways.



What is Human-Computer Interaction (HCI)?

Human–computer interaction (HCI) is a multidisciplinary field of study focusing on the design of computer technology and, in particular, the interaction between humans (the users) and computers. While initially concerned with computers, HCI has since expanded to cover almost all forms of information technology design.

https://en.wikipedia.org/wiki/Human%E2%80%93computer_interaction

<https://www.interaction-design.org/literature/topics/human-computer-interaction>

Multi- and inter-disciplinary

- Cognitive Science
- Computer Science
- Disability Studies
- Electrical Engineering
- Education
- Industrial & Visual Design
- Health Informatics
- Human Factors Engineering
- Media Studies
- Social Psychology
- Software Engineering
- Science, Technology, & Society

Multi- and inter-disciplinary

“It no longer makes sense to regard HCI as a specialty of computer science; HCI has grown to be broader, larger and much more diverse than computer science itself. HCI expanded from its initial focus on individual and generic user behavior to include social and organizational computing, accessibility for the elderly, the cognitively and physically impaired, and for all people, and for the widest possible spectrum of human experiences and activities.” -John M. Carroll

<https://www.interaction-design.org/literature/topics/human-computer-interaction>

HCI versus HCC

Human-Computer Interaction vs. Human-Centered Computing

Synopsis

Human-Centered Computing (HCC) supports research in human-computer interaction (HCI), taken broadly, integrating knowledge across disciplines—such as the social and behavioral

- They effectively the same thing
- HCC is probably more inclusive
 - Computing can be human-centered without requiring interaction
 - NSF recently switched the name of their call to HCC
- By in large, the research community uses HCI, so I'll default to that
 - A few programs use HCC (GT, UMBC, Clemson) or Human-Centered Design (UW)

<https://beta.nsf.gov/funding/opportunities/iis-human-centered-computing-hcc>

HCI versus UX

Human-Computer Interaction vs. User Experience

- HCI tends to be more academic, UX more industry-focused
- Improving UX is one of many goals that HCI researchers have
 - More on research contribution styles next class

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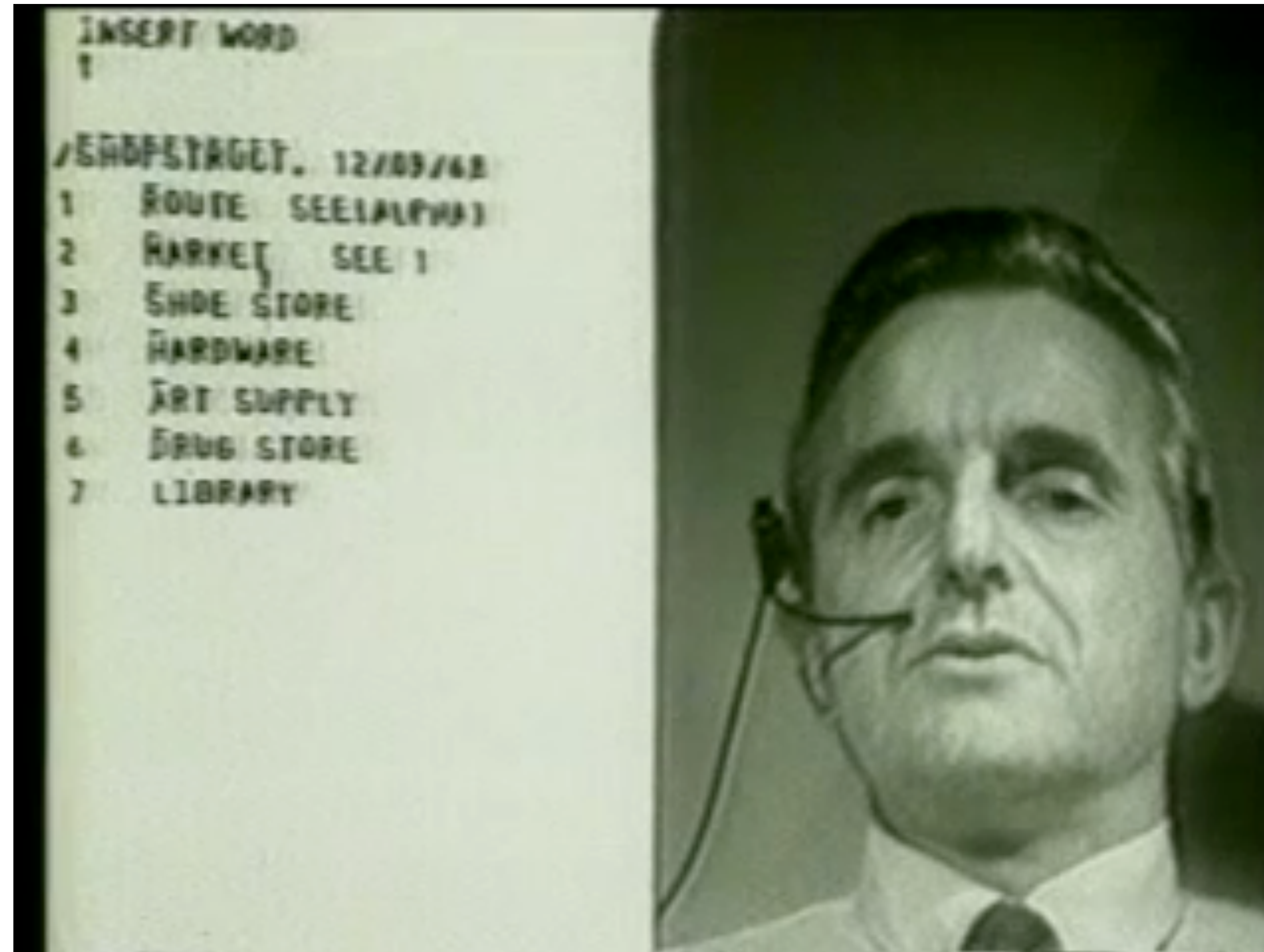
HCI History

- This is a very quick history lesson.
- My background is in Computer Science, which influences the lens I use

Pre-History

- People have aimed to improve how people interact with computers ever since they have existed
- One example: Engelbart's NLS

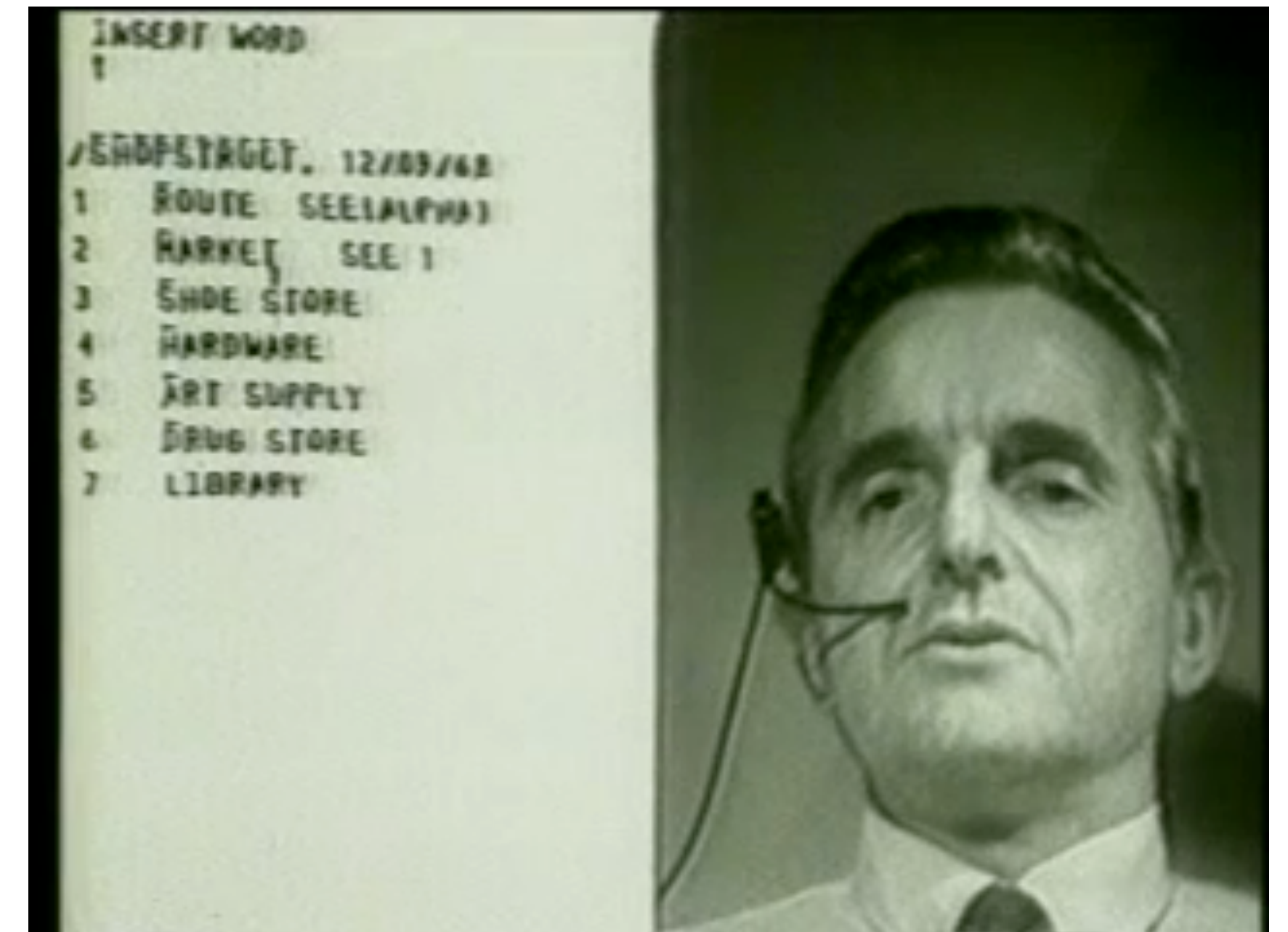
Doug Engelbart's NLS (1968)



<http://www.douengelbart.org/firsts/1968-demo-interactive.html>

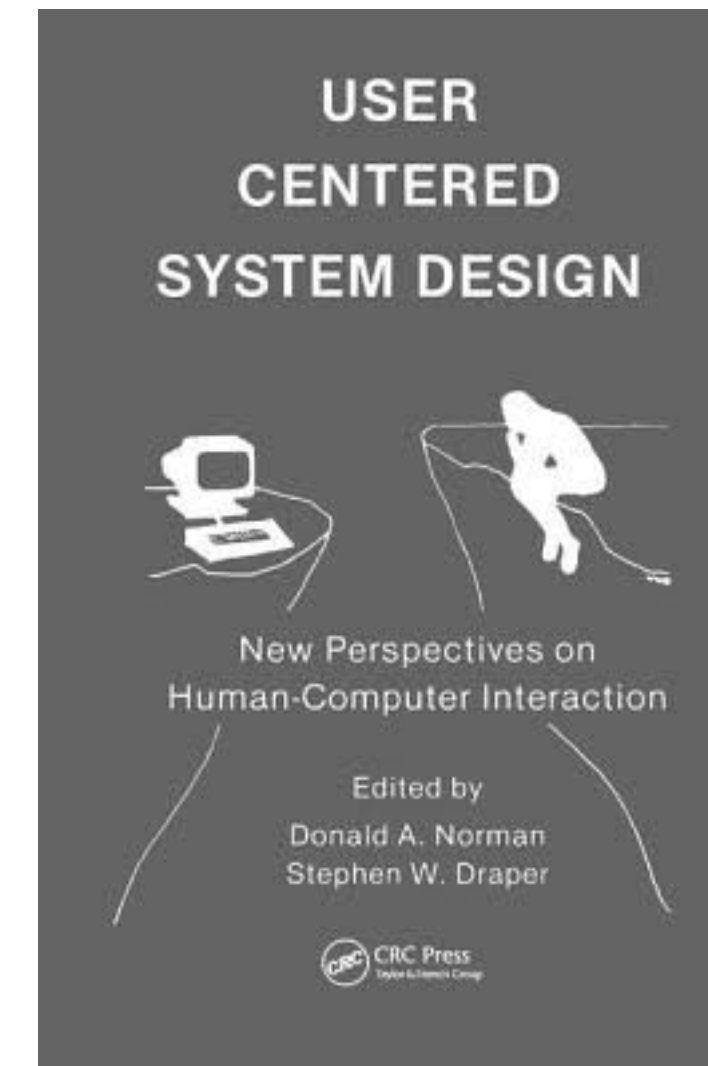
Doug Engelbart's NLS (1968)

- Direct linking between pages
- Invention of the mouse
- Simple graphics
- A chording keyboard
- Remote collaboration
- Others thought it was irrelevant because “the terminal can do the same”



Pre-History

- Early 1980's, idea that computers should be usable by *everyone* began to emerge
- Principles like learnability, efficiency, safety, etc. were beginning to be articulated
- Research community on the topic began coalescing



CHI 1982

- CHI: ACM Conference on Human Factors in Computing Systems
- ACM: Association for Computer Machinery
 - Driven by computer science in the early years
- Human Factors
 - Engineering and design of products, with an eye towards ergonomics, safety, and reducing error



Gaithersburg, MD
Pioneering researcher
Alphonse Chapanis (top right) spoke
at the first conference, which was initiated
by Ben Shneiderman (bottom center) and Bill
Curtis (top left, with Jerry Weinberg).

CHI 1982

- CHI turned 40 in 2022
- Is still the flagship publication venue for HCI research
- CHI -> HCI
 - We moved the “human” first



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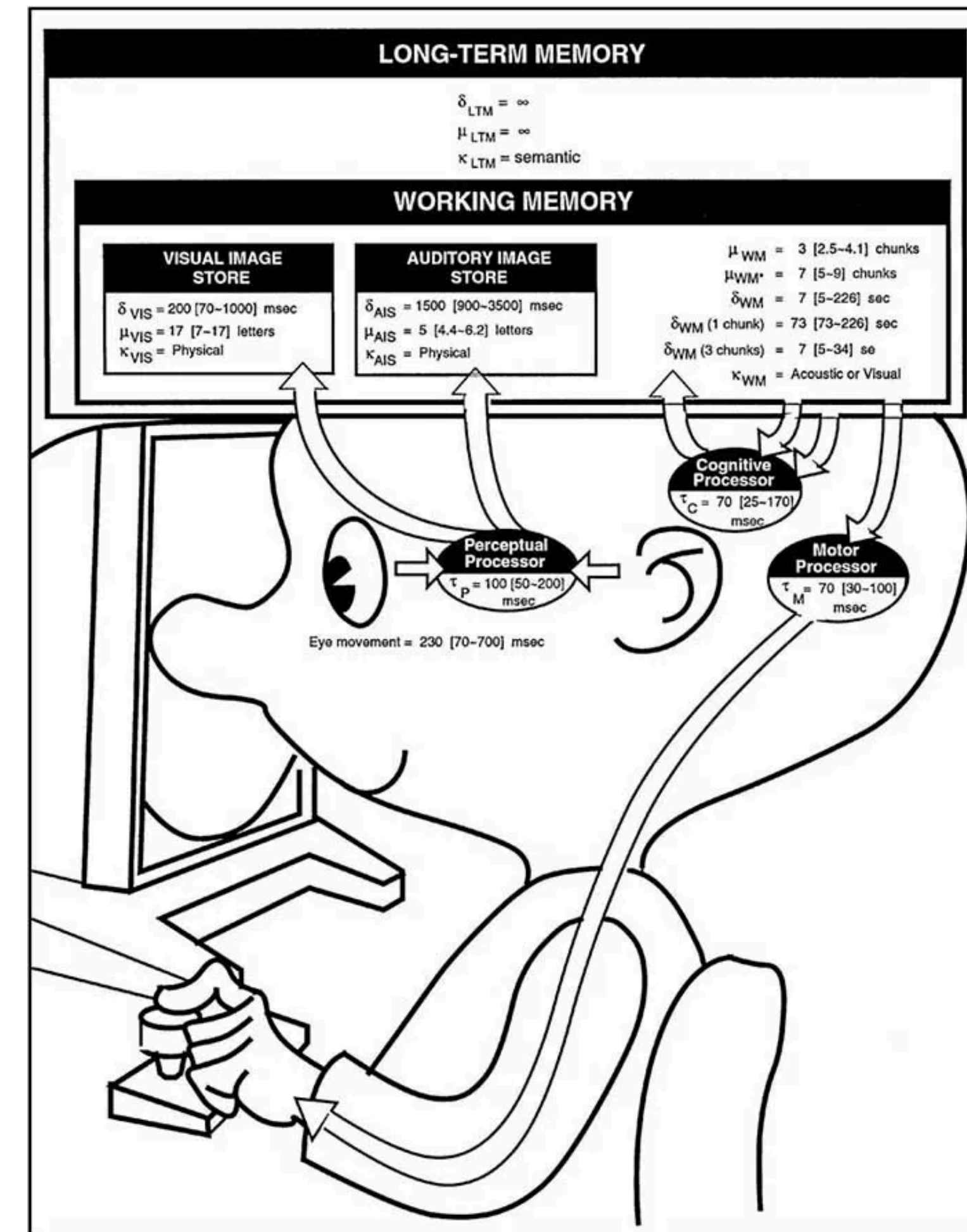
Three Waves of HCI Research

- First Wave ('80s): Cognitive processes and ergonomic perspectives
 - Second Wave ('90s and '00s): Groups working with applications
 - Third Wave ('10s): Computing in Everyday Life
-
- I've also seen these separated as the Personal Computing, Groupware, and Ubiquitous Computing eras

<https://dl.acm.org/doi/10.1145/2804405>

First Wave ('80s)

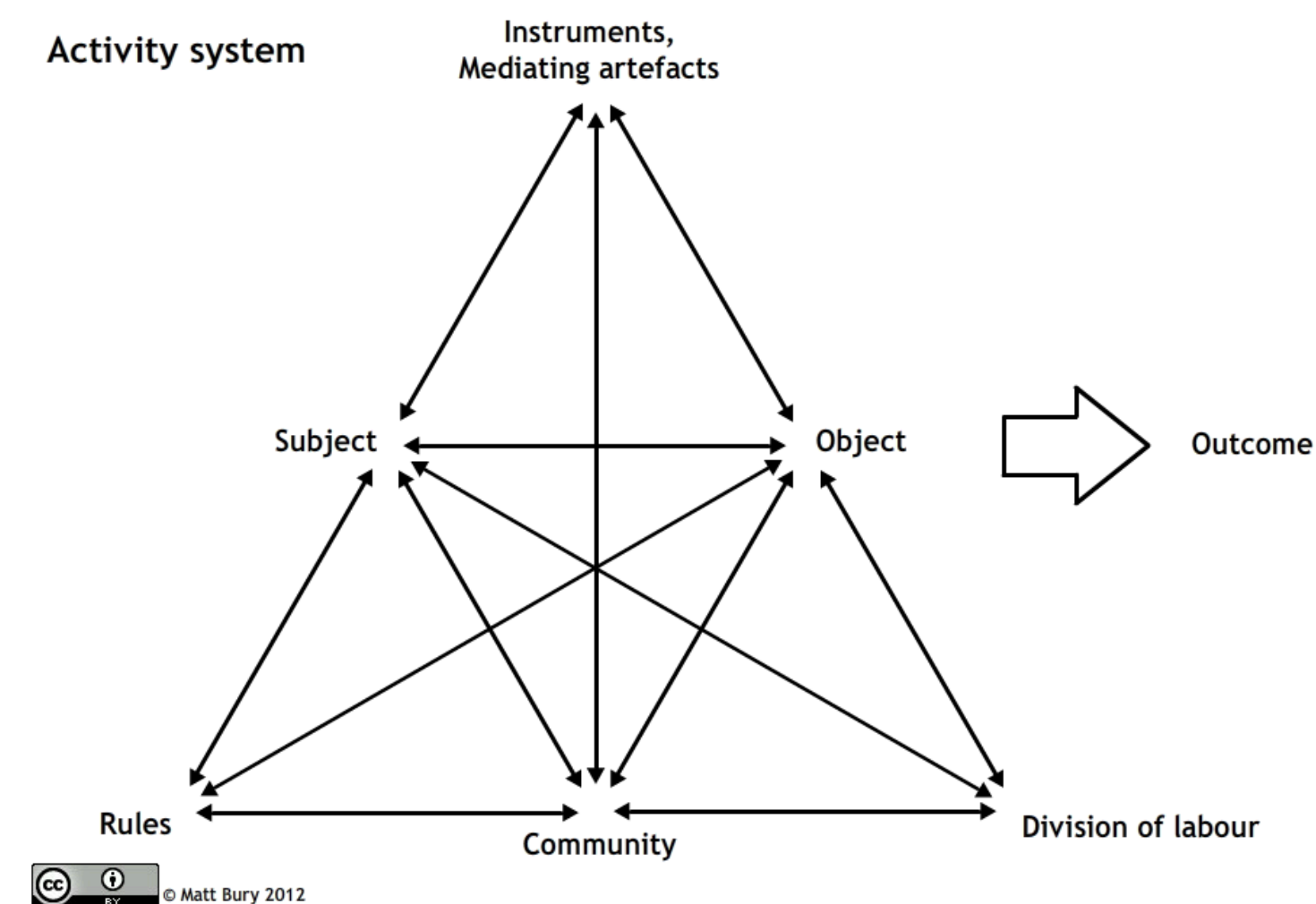
- A person's cognitive processes could be modeled and predicted, enabling software designers to evaluate and improve interface designs
- GOMS Model (Goals, Operators, Methods, Selection rules)
 - A method used to predict where a person might have trouble with an interface



<https://www.amazon.com/Psychology-Human-Computer-Interaction-Stuart-Card/dp/0898598591>

Second Wave ('90s and '00s)

- Groups working collectively
- Introduction of social psychological theories
 - Distributed cognition: developing shared understanding and action among multiple people
 - Activity theory: considering an individual's practice in the context of the community and environment that they are in



CHI Growth



The screenshot shows the CHI 2009 website with the header "CHI 2009 DIGITAL LIFE NEW WORLD". The navigation bar includes links for WELCOME, ATTENDING, AUTHORS, SPONSORS, EXHIBITORS, ORGANIZERS, and PRESS. Below the navigation bar, there is a purple banner with links for "Call for papers", "Format", "Guides", "Present", and "Madness". The main content area features a large image of a person using a stylus on a tablet. Below the image, the word "AUTHORS" is displayed in blue. The section is titled "Subcommittees" in bold blue text. The text below explains that CHI 2009 anticipates over 1,000 papers and notes, and that the review process is managed by topical subcommittees to improve review quality. Authors are encouraged to select the best subcommittee for their submission.

AUTHORS

Subcommittees

CHI 2009 anticipates submission of over 1,000 Papers and Notes. The review process needs to handle this load while also improving the quality of reviews. The organization of the CHI program committee into topical subcommittees helps achieve this by having you, the author, select the best subcommittee to review your submission.

<https://web.archive.org/web/20090321195157/http://chi2009.org/Authors/CallForPapers/Subcommittees.html>

CHI Growth

Interaction Beyond the Individual

This subcommittee will focus on papers which consider aspects of interaction which extend beyond a single user. These contributions will be judged in part by their extension of knowledge about large and small groups of people's interaction with technology and with each other through technology and/or by their innovation in creating new systems or techniques to support these interactions.

Subcommittee

Chair: [Wendy Kellogg](#) (IBM)

Associate Chairs:

- Mark Ackerman (University of Michigan)
- Barry Brown (UC San Diego)
- Scott Counts (Microsoft Research)
- Susan Fussell (Cornell)
- Darren Gergle (Northwestern)
- Sara Kiesler (CMU)
- Cliff Lampe (Michigan State University)
- Gloria Mark (UC Irvine)
- Judy Olson (UC Irvine)

Behavioral Study and Theory

This subcommittee will focus on papers which contribute improved understanding, measures, or models of people and/or context which can be applied to address HCI problems. These contributions will be judged in part by their extension of our basic understanding **of human behavior and/or their context of activity** and the practical impact this may have on HCI practice and research.

Subcommittee

Chairs:

[Geraldine Fitzpatrick](#) (University of Sussex)

[Carl Gutwin](#) (University of Saskatchewan)

Associate Chairs:

- Anthony Hornof (University of Oregon)
- Dave Kirk (Microsoft Research)
- Bonnie Nardi (University of California, Irvine)
- Stephen Payne (Bath University)
- Mark Rouncefield (Lancaster University)
- Louise Barkhuus (University of California, San Diego)
- Kenton O'Hara (HxI Initiative, Australia)

<https://web.archive.org/web/20090321195157/http://chi2009.org/Authors/CallForPapers/Subcommittees.html>

CHI Growth

- In making a subcommittee choice you should make careful consideration of **what the most central and salient contribution of your work is**, even if there are several different contributions... Should it be evaluated in terms of the **usage context for the target user community? The novel methodology developed for your study? The system and interaction techniques you have developed?**... Even if there are several subcommittees that could offer fair and expert assessments of this work, go with the one that really fits the most important and novel contributions of your paper.

<https://web.archive.org/web/20090321195157/http://chi2009.org/Authors/CallForPapers/Subcommittees.html>

Third Wave ('10s)

- Computing blends professional and personal lives
- Evolution spurred on by changes in the kinds of computers people use
- Research begins examining computing use beyond productivity
 - Entertainment
 - Health
 - Education



CHI Growth, continued

2010	Atlanta	 USA	http://www.chi2010.org	2,384
2011	Vancouver	 Canada	http://www.chi2011.org	2,822
2012	Austin, Texas	 USA	http://chi2012.acm.org	2,616
2013	Paris	 France	http://chi2013.acm.org	3,443
2014	Toronto	 Canada	http://chi2014.acm.org	3,001
2015	Seoul	 South Korea	http://chi2015.acm.org	2,896
2016	San Jose	 USA	http://chi2016.acm.org/	3,624
2017	Denver	 USA	https://chi2017.acm.org	2,939
2018	Montreal	 Canada	https://chi2018.acm.org/	3,372
2019	Glasgow	 UK	https://chi2019.acm.org/	3,855
2020	Cancelled ^[a]		https://chi2020.acm.org/	Canceled due to Covid-19 ^[13]
2021	Virtual ^[b]		https://chi2021.acm.org/	5,145
2022	New Orleans, LA	 USA	https://chi2022.acm.org/	3,806
2023	Hamburg	 Germany	https://chi2023.acm.org/	4,721
2024	Honolulu	 USA	https://chi2024.acm.org/	4,120
2025	Yokohama	 Japan	https://chi2025.acm.org/	5,675

https://en.wikipedia.org/wiki/Conference_on_Human_Factors_in_Computing_Systems

In Parallel Worlds...

- Emergence of researchers examining the creation and consequences of technology
 - Science, Technology, and Society (STS)
- Increasing technology adoption in application areas
 - Health
 - Learning sciences
- Regulation of industry to support the rights of individuals
 - Americans with Disabilities Act (ADA)
 - Privacy law

Fourth Wave? ('20s)

Critical computing?

- Deeper reflection on computing and society
 - Who we are designing technology to support (wealthy people, corporations)
 - When technology is causing harm rather than benefit
- It could be argued that this wave isn't new
 - STS has been doing it for decades
 - But its uptake in HCI is relatively new, and spreading rapidly

Fourth Wave? ('20s)

AI?

- What computing “is” is changing
 - Algorithmically-generated content
 - Algorithmic personalization
- It could be argued that this wave isn't new
 - Social media feeds have been around for a decade+
 - NLP and Computer Vision have researched text/image generation for decades

HCI happens outside of CHI

- SIGCHI: ACM's Special Interest Group for HCI
 - The organizational body that sponsors HCI conferences
- CHI is useful for looking at HCI research because *all of* CHI is HCI
 - Other venues publish a significant amount of HCI literature, with no or little difference in quality

☰ Upcoming Conferences

📅 Calendar

GROUP 2025

South Carolina, USA · Jan 12, 2025 – Jan 15, 2025

The ACM International Conference on Supporting Group Work (GROUP) is a premier yet intimate and welcoming venue for research on Computer Supported Cooperative Work, Sociotechnical Studies, Practice-Centered Computing, Human Computer Interaction,...

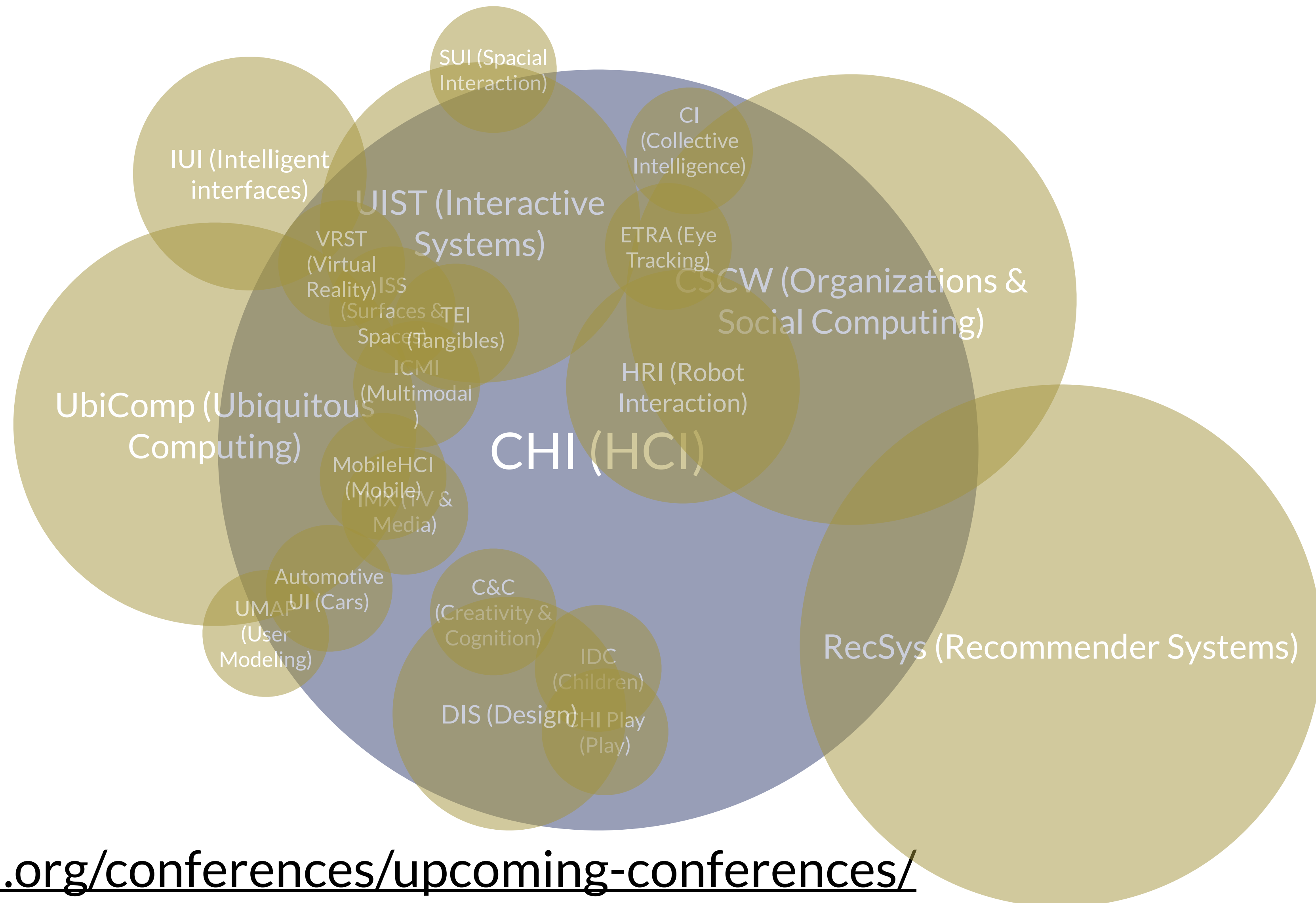
TEI 2025

Bordeaux, France · Mar 03, 2025 – Mar 06, 2025

TEI'25 is the 19th annual conference dedicated to presenting the latest results in tangible, embedded, and embodied interaction. The ACM TEI conference has gained substantial visibility and activity over the past decade, resulting in outstanding works from past...

<https://sigchi.org/conferences/upcoming/>

HCI happens outside of CHI



<https://sigchi.org/conferences/upcoming-conferences/>

Today's class

- Defining Human-Computer Interaction (HCI)
- History of HCI
- Course overview

Course Overview

- Learning objectives
- Who I am
- Staying in touch
- Structure & readings
- Assignments
- Grading
- Website overview

Learning Objectives

- By the end of this course, you should be able to:
 - Analyze the motivation and contribution of any Human-Computer Interaction research paper, describing how it extends the field of literature.
 - Articulate how Human-Computer Interaction relates to other areas of research, both within Informatics and in academia at large.
 - Develop proposals for research projects which develop new knowledge to one or more areas of Human-Computer Interaction.
 - Appreciate different methods of inquiry, topics, and objectives used in different Human-Computer Interaction research areas.

Learning Objectives

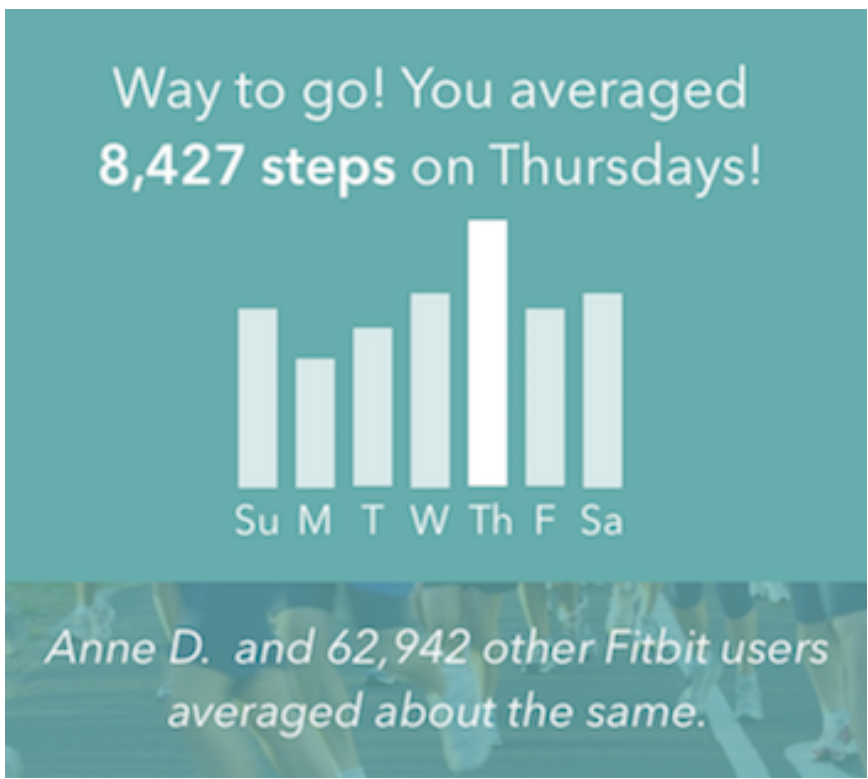
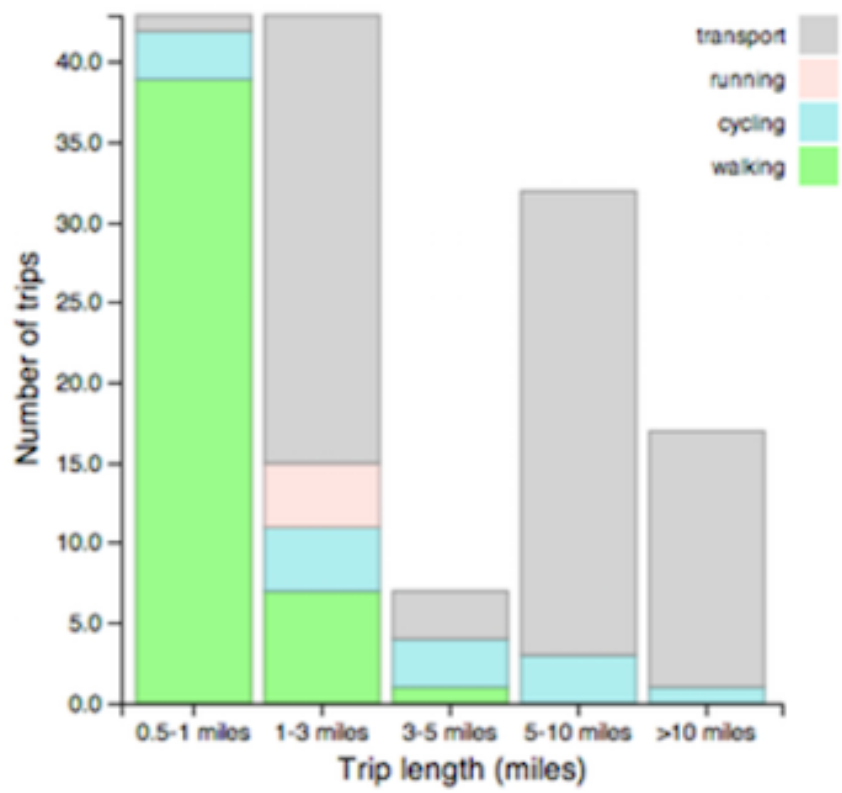
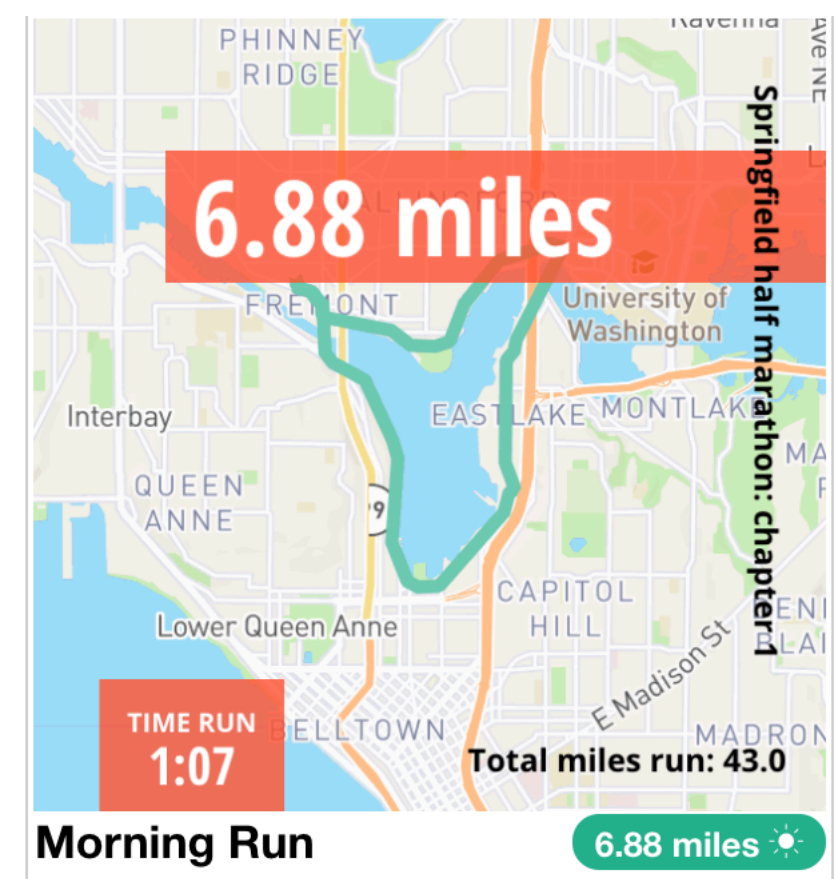
- This is *not* a course on methods used in HCI research
 - Those are largely covered by Qualitative (203) and Quantitative (205) Methods
- This is *not* a course on applying HCI principles to the design of software, apps, etc. in industry
 - That is largely covered by User Interface Design (231)
- Instead, this class aims to give background on what contributions are core to HCI research and how to motivate them in writing
 - This class prioritizes breadth (more topics) over depth
 - CSCW (251), Ubicomp (241), and many special topics (295) provide greater depth

Who I am

- Ph.D. Computer Science & Engineering, University of Washington 2018
- B.S. Computer Science, University of Virginia 2012
- Joined UCI Informatics in 2018, tenured in 2024
- Internships at Microsoft & Adobe, collaborations with Snap(chat) & NAVER



Who I am



Who I am

- I consider HCI my primary research area
- I regularly publish at and review for HCI venues
 - CHI, Ubicomp/IMWUT, and CSCW are my “main” venues
- Within HCI, I focus on Health, Ubiquitous Computing, and Social Computing



Around the room!
Say your name, program & year,
and something fun about you

Staying in touch

- Web: <http://inf232-wi26.depstein.net/>
- Email me: epstein@ics.uci.edu
- Office hours: generally by appointment (send me an email)

Structure & Readings

- In-person, synchronous, no recording, no hybrid
 - Discussion is a key component of the class
 - Quality hybrid discussions are hard to pull off
- Let me know if you need to miss a class
 - I expect many of you will need to miss a few
 - If you miss a week or less, your grade won't be impacted

Structure & Readings

- Topics will be organized by CHI's subcommittees
- There are not enough weeks to cover each subcommittee
- Poll is up to help determine which subcommittees to focus on (complete it by Friday)

List of the Subcommittees

Sixteen subcommittees are listed and described below. Each has a title, short description, and an indication of who will Chair and serve on the subcommittee and if a subcommittee consists of multiple tracks. Subcommittees have been constructed with an eye to maintaining logically coherent clusters of topics.

- [User Experience and Usability](#)
- [Specific Applications Areas](#)
- [Learning, Education, and Families](#)
- [Interaction Beyond the Individual](#)
- [Games and Play](#)
- [Privacy and Security](#)
- [Visualization](#)
- [Health](#)
- [Accessibility and Aging](#)
- [Design](#)
- [Building Devices: Hardware, Materials, and Fabrication](#)
- [Interacting with Devices: Interaction Techniques & Modalities](#)
- [Blending Interaction: Engineering Interactive Systems & Tools](#)
- [Understanding People: Theory, Concepts, and Methods](#)
- [Critical Computing, Sustainability, and Social Justice](#)
- [Computational Interaction](#)

Structure & Readings

- Each week will cover a different subcommittee/topic at CHI
- First day: an overview of the topic
 - What disciplines it draws from
 - Core problems or research questions
 - Comparison between HCI research on the topic and research in related communities
- Second day: discussion of recent papers on that topic
 - What contributions are made and how they are made
 - What makes a problem on this topic worthy of study

Structure & Readings

- “Overview Day” readings:
 - *Calls for papers* from the CHI subcommittee and related publication venues
 - *Framing* papers which introduce a concept or approach commonly used
- “Discussion Day” readings:
 - *Recent contributions* which will be used to better understand the topic
- Doing the reading is a major component of the course
 - No exams, no quarter-long project

Structure & Readings

- Going to use Perusall, a collaborative reading/commenting tool
- Comments graded binary 0/1 to simplify grading, but more and deeper comments will lead to better discussion and thus better learning

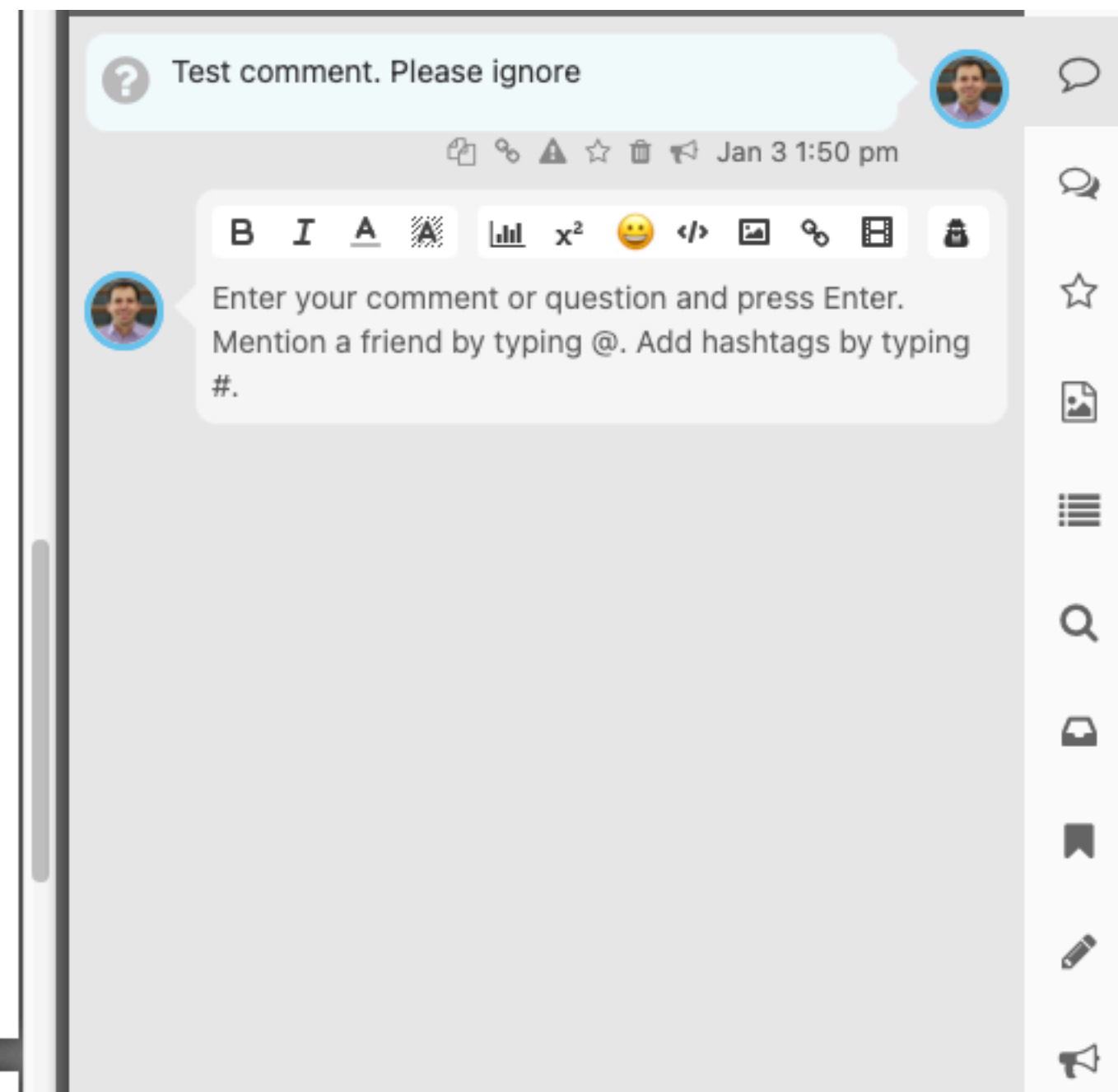
Insights

- Knowledge generated by HCI research can be categorized into certain contribution types.
- Each contribution type has key characteristics that imply how it is judged.
- The contribution types used for submissions to the CHI conference have evolved over time to distill types of knowledge from other concerns.

All scholarly fields strive to contribute new knowledge. In the field of human-computer interaction (HCI), this new knowledge increasingly comes in rich forms like videos and demos, but the archival research paper remains the most widely used and accepted capture and delivery mechanism for research knowledge. The knowledge contribution made by a research paper—or more precisely, made by the work a research paper describes—is any research paper's central feature. For example, a theoretical physics paper may contribute a new mathematical model for the behavior of light near black holes. A civil

engineering paper may contribute a new method for stress-testing bridges. A social anthropology paper may contribute an account of people's reactions to teen pregnancies in rural religious communities. Whatever the field of inquiry, whatever the phenomenon of interest, every research paper strives to make a research contribution by offering new knowledge. In an effort to distinguish this kind of knowledge from everyday know-how, some scholars even capitalize the term: Knowledge.

In the whole of human inquiry, there are, of course, countless specific research contributions to be made. But



Structure & Readings

- Each subcommittee will have 1-2 of you in charge of leading discussion
- You will present on each week's readings
 - On “Overview Day”, providing some additional background on the topic and its history in HCI
 - On “Discussion Day”, picking the recent papers and being the expert on their motivation and takeaways

Structure & Readings

- Read smart
 - No need to read the formatting expectations, submission method etc. in Calls for Papers, unless you're curious
 - Can skip critique of recent papers' methods, someone else reviewed them and felt they were conducted rigorously enough
 - Focus on what's relevant today from the "old" readings, not just that they're "old"
- There are required readings for Thursday
- Again, fill out the discussion leading preferences poll ASAP

Structure & Readings

- At some point during the quarter, my partner and I will have a baby!
- Due date is in week 9, but could be sooner or later
- Once that happens, I will largely be offline
- I am lining up guest lectures for weeks 9 and 10, and back-ups for week 8
- But, expect that there will be disruption



CHI 2026



- Paper decisions are happening shortly
- It's unlikely that the accepted papers will be available in time for us to read them in class
- But, maybe we'll recruit a few UCI authors to come in to share their work later in the quarter

Assignments

- 3 assignments, giving practical experience assessing and producing motivations for HCI contributions
 - Reverse outline (individual): Pick two papers, deconstruct its motivation, contribution, and methods
 - Research proposals (groups, 2x): Write proposals for two different “calls” at the intersection of different subcommittees

Grading

- 45% Assignments
 - 15% Reverse Outline, 15% Proposal 1, 15% Proposal 2
- 35% Readings
 - 25% Reading Reports, 10% Leading Discussion
- 20% Participation
 - Attendance (on-time!) is expected, and there will be penalty for many absences
 - I strongly encourage speaking up in class, and we will have many small group discussions to facilitate other forms of participation

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Website Overview

INF 232 - Research in HCI - Winter 2025

[Home](#) | [Assignments](#) | [Readings](#) | [Calendar](#)

Basic Information

Class Time: TuTh 12:30-1:50pm

Class Location: ICS 180

Perusall:

<https://app.perusall.com/courses/in4matx-232-winter-2026/>

Canvas:

<https://canvas.eee.uci.edu/courses/79168>

Email: epstein@ics.uci.edu

Office hours: By appointment

Course Staff

Professor Daniel Epstein

Welcome! This class focuses on research topics in Human-Computer Interaction (HCI), focused on developing appreciation for and expertise in different kinds of research contributions in HCI. HCI spans over four decades as a research field, evolving from understanding and improving computer use in traditional workplace settings to developing, studying, and critiquing the wide range of computing devices that people use in their everyday lives.

This course will require a substantial amount of reading, as well as analysis of literature and writing. While there will be a few lectures dedicated to giving background on the field, in most weeks we will instead look to understand the research literature through discussion.

We will emphasize open discussion and feedback in all aspects of the course.

<https://inf232-wi26.depstein.net/>

Final Thoughts

- Since this class is required for the Informatics PhD, and many of you are in other programs, your interests are diverse
 - Some of you self-identify as HCI researchers, others do not
 - But I believe that all of you can connect your research interests to HCI, if you wish to
 - My goal is to help you find your connection, or broaden your background if you already have one

Final Thoughts

- As much as I can, I will share what I know about HCI
 - I think I have a relatively broad understanding of the HCI literature and community
 - But there are plenty of areas where I have limited knowledge
 - Please establish your expertise when you have it, disagree with me, etc.
- You'll get out of this class what you put in
 - You can pass the class by minimally engaging with the readings and not asking questions during class, but you'll learn less
 - Again, I'm here to share what I know and provide some structure for you to learn

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History & Course Overview

Daniel Epstein